$\boxed{4.10}$

a) $Z(T,N,V,B) = \sum_{\{s\}} e^{-\beta \mathcal{H}} = \sum_{\{s\}} e^{-\beta\left(\sum_i \frac{p_i^2}{2m} - \mu B s_i^z\right)}$

From last homework we know integration over $d^3\vec{q}_i$ and $d^3\vec{p}_i$ will give us $\frac{V^N}{N!\,\lambda^{3N}}$ where $\lambda = \frac{h}{\sqrt{2\pi m k T}}$

So we just have spins. For one molecule

$$Z_1 = \sum_{\{s\}} e^{+\beta\mu B s_s} \quad \text{where } \{s\} \text{ is } +1, 0, -1$$

$$Z_1 = e^{\beta\mu B} + 1 + e^{-\beta\mu B} = 1 + 2\cosh(\beta\mu B)$$

With N spins $\quad Z = \frac{V^N}{N!\,\lambda^{3N}}\left(1 + 2\cosh(\beta\mu B)\right)^N$

b) $P(1) = \frac{e^{\beta\mu B}}{Z_1} = \frac{e^{\beta\mu B}}{1+2\cosh(\beta\mu B)}$

$P(0) = \frac{1}{1+2\cosh(\beta\mu B)}$ $\qquad$ Use Z_1 since this is for a specific molecule

$P(-1) = \frac{e^{-\beta\mu B}}{1+2\cosh(\beta\mu B}$

c) $M = \mu \sum s_i^z \qquad \langle M\rangle = \mu\langle \sum s_i^z\rangle = \mu N \langle s_i^z\rangle$

$$\langle s_i^z\rangle = \frac{\sum_{\{s\}} s_s^z e^{+\beta\mu B s_s^z}}{1+2\cosh(\beta\mu B)} = \frac{e^{+\beta\mu B} + 0 - e^{-\beta\mu B}}{1+2\cosh(\beta\mu B)} = \frac{2\sinh(\beta\mu B)}{1+2\cosh(\beta\mu B)}$$

$$\frac{\langle M\rangle}{V} = \frac{\mu N}{V}\,\frac{2\sinh(\beta\mu B)}{1+2\cosh(\beta\mu B)}$$

$\boxed{4.10}$ cont...

d) $\chi = \left.\dfrac{\partial \langle M \rangle}{\partial B}\right|_{B=0}$

$$\frac{\partial \langle M \rangle}{\partial B} = \mu N \left[\frac{2\mu\beta \cosh(\beta\mu B)}{1+2\cosh(\beta\mu B)} + \frac{2\sinh(\beta\mu B)(-1)\, 2\mu\beta \sinh(\beta\mu B)}{(1+2\cosh(\beta\mu B))^2} \right]$$

$$= \frac{2\mu^2 N\beta}{} \left[\frac{\cosh(\beta\mu B) + 2\cosh^2(\beta\mu B) - 2\sinh^2(\beta\mu B)}{(1+2\cosh(\beta\mu B))^2} \right]$$

$$= \frac{2\mu^2 N\beta}{} \left[\frac{2 + \cosh(\beta\mu B)}{(1+2\cosh(\beta\mu B))^2} \right]$$

Now evaluate @ $B=0$

$$\left.\frac{\partial \langle M \rangle}{\partial B}\right|_{B=0} = \frac{2\mu^2 N\beta}{} \left(\frac{3}{9} \right) = \frac{2\mu^2 N\beta}{3}$$

$\boxed{4.12}$ $\mathcal{H}_{rot} = \frac{1}{2I}\left(p_\theta^2 + \frac{p_\varphi^2}{\sin^2\theta}\right) - \mu E\cos\theta$

$Z_{rot} = \sum e^{-\beta \mathcal{H}_{rot}}$ Transform sum to integral

$$Z_{rot} = \frac{1}{h^2}\int_0^\pi d\theta \underbrace{\int_0^{2\pi} d\varphi}_{2\pi} \int_{-\infty}^{\infty} dp_\theta \int_{-\infty}^{\infty} dp_\varphi \exp\left\{\frac{-\beta}{2I}p_\theta^2\right\}\exp\left\{\frac{-\beta}{2I}\frac{p_\varphi^2}{\sin^2\theta}\right\} e^{\beta\mu E\cos\theta}$$

$$\int_{-\infty}^{\infty} e^{-\frac{\beta p_\theta^2}{2I}} = \sqrt{\frac{2\pi I}{\beta}}$$

So $Z_{rot} = \frac{2\pi}{h^2}\sqrt{\frac{2\pi I}{\beta}}\int_0^\pi d\theta \int_{-\infty}^{\infty} dp_\varphi\, e^{\frac{-\beta p_\varphi^2}{2I\sin^2\theta}} e^{\beta\mu E\cos\theta}$

$$\int_{-\infty}^{\infty} dp_\varphi\, e^{\frac{-\beta p_\varphi^2}{2I\sin^2\theta}} = \sqrt{\frac{2\pi I\sin^2\theta}{\beta}}$$

Now $Z_{rot} = \frac{2\pi}{h^2}\frac{2\pi I}{\beta}\int_0^\pi \sin\theta\, e^{\beta\mu E\cos\theta}\, d\theta$

$u = \beta\mu E\cos\theta$

$du = -\beta\mu E\sin\theta\, d\theta$

$$Z_{rot} = \left(\frac{2\pi}{h}\right)^2\frac{I}{\beta}\left(\frac{-1}{\beta\mu E}\right)\int_{\beta\mu E}^{-\beta\mu E} du\, e^u = -\left(\frac{2\pi}{\beta h}\right)^2\frac{I}{\mu E}\left(e^{-\beta\mu E} - e^{\beta\mu E}\right)$$

$$Z_{rot} = \left(\frac{2\pi}{\beta h}\right)^2\frac{I}{\mu E}\, 2\sinh(\beta\mu E)$$

b) $P = \mu\langle\cos\theta\rangle$

$$\langle\cos\theta\rangle = \frac{1}{Z_{rot}}\int_0^\pi d\theta\int_0^{2\pi} d\varphi\int_{-\infty}^{\infty} dp_\theta\int_{-\infty}^{\infty} dp_\varphi \cos\theta\, e^{-\beta\mathcal{H}_{rot}}$$

This is the same as problem a) until we get to θ integration

$$\langle\cos\theta\rangle = \frac{1}{Z_{rot}}\left(\frac{2\pi}{h}\right)^2\frac{I}{\beta}\int_0^\pi \sin\theta\cos\theta\, e^{\beta\mu E\cos\theta}\, d\theta$$

$u = \beta\mu E\cos\theta$ $\quad du = -\beta\mu E\sin\theta\, d\theta$ $\quad \cos\theta = \frac{u}{\beta\mu E}$

$$\langle\cos\theta\rangle = \frac{-1}{Z_{rot}}\left(\frac{2\pi}{\beta h}\right)^2\frac{I}{(\mu E)^2}\int_{\beta\mu E}^{-\beta\mu E} u e^u\, du$$

Use integration by parts

$\boxed{4.12}$ cont...

$$\langle \cos\theta \rangle = \frac{-1}{Z_{rot}} \left(\frac{2\pi}{h\beta\mu E}\right)^2 \frac{I}{\beta}\left(-\beta\mu E\, 2\cosh(\beta\mu E) + 2\sinh(\beta\mu E)\right)$$

$$P = \mu\langle\cos\theta\rangle = \frac{\cancel{2}\cancel{I}\cancel{\mu}}{\beta}\left(\frac{2\pi}{\cancel{h}\beta\cancel{\mu}E}\right)^2\left(\beta\mu E\cosh(\beta\mu E) - \sinh(\beta\mu E)\right)\frac{\cancel{\beta^2}\cancel{h^2}}{\cancel{(2\pi)^2}}\frac{\cancel{\mu}E}{\cancel{I}\cancel{2}\sinh(\beta\mu E)}$$

$$P = \frac{1}{\beta E\sinh(\beta\mu E)}\left[\beta\mu E\cosh(\beta\mu E) - \sinh(\beta\mu E)\right]$$

$$P = \mu\coth(\beta\mu E) - \frac{1}{\beta E}$$

c) $$\chi_T = \frac{\partial P}{\partial E}\Big|_{E=0} = \frac{-\mu^2\beta}{\sinh^2(\beta\mu E)} + \frac{1}{\beta E^2}\Big|_{E=0}$$

But that is infinite, so lets combine to one term and expand $\beta\mu E$

let $x = \beta\mu E$ $\quad \chi_T \simeq \dfrac{-x^2 + \sinh^2 x}{\frac{x^2}{\mu^2\beta}\sinh^2 x}$ $\qquad \sinh x \sim x + \frac{x^3}{6}$

$$\chi_T = \frac{\left(-x^2 + x^2 + \frac{x^6}{36} + \frac{x^4}{3}\right)\mu^2\beta}{\left(x^4 + \frac{x^8}{36} + \frac{x^6}{3}\right)} = \frac{\mu^2\beta\left(\frac{x^6}{36} + \frac{x^4}{3}\right)}{\left(x^4 + \frac{x^8}{36} + \frac{x^6}{3}\right)}$$

If we only keep lowest order remaing terms (x^4) we get

$$\chi_T = \mu^2\beta\left(\frac{x^4}{3}\right)\cdot\frac{1}{x^4} = \frac{1}{3}\beta\mu^2 = \frac{\mu^2}{3kT} \quad \text{Curie Law}$$

d) $$E_{rot} = \langle U\rangle = -\frac{\partial}{\partial\beta}\ln Z_{rot} = -\frac{\partial}{\partial\beta}\left[\ln\left((2\pi)^2 2I\sinh(\beta\mu E)\right) - \ln\left(\beta^2 h^2 \mu E\right)\right]$$

$$\langle U\rangle = \frac{-\mu E(2\pi)^2 2I\cosh(\beta\mu E)}{(2\pi)^2 2I\sinh(\beta\mu E)} + \frac{2\beta h^2\mu E}{\beta^2 h^2 \mu E}$$

$$\langle U\rangle = -\mu E\frac{\cosh(\beta\mu E)}{\sinh(\beta\mu E)} + \frac{2}{\beta} = 2kT - \mu E\coth(\beta\mu E)$$

$\boxed{4.12}$ cont...

d) cont...

$$\langle u \rangle = 2kT - \mu E \coth(\beta \mu E) = 2kT - \mu E \coth\left(\frac{\mu E}{kT}\right)$$

For low T $\frac{\mu E}{kT} \to \infty$ $\quad \coth\left(\frac{\mu E}{kT} \to \infty\right) = 1$

$$\langle u \rangle \simeq 2kT - \mu E$$

For high T $\frac{\mu E}{kT} \to 0$ $\quad \coth x = \frac{\cosh x}{\sinh x} \sim \frac{1}{x}$

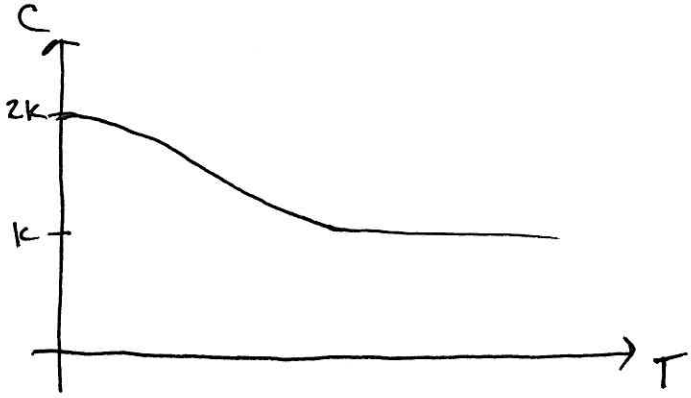
$$\langle u \rangle = 2kT - \mu E \frac{kT}{\mu E} = kT$$

2 degrees of freedom giving $\frac{1}{2}kT$

e) $$C = \frac{\partial \langle u \rangle}{\partial T} = 2k - \frac{\mu^2 E^2}{kT^2} \frac{1}{\sinh^2\left(\frac{\mu E}{kT}\right)}$$

Large T $\quad \sinh^2\left(\frac{\mu E}{kT}\right) \sim \left(\frac{\mu E}{kT}\right)^2$ $\quad C = 2k - \frac{\mu^2 E^2}{kT^2} \frac{k^2 T^2}{\mu^2 E^2}$

$$C \sim 2k - k = k$$

Small T: $\sinh^2$ goes to ∞ faster than $T^2 \to 0$

so $\quad C \sim 2k - 0 = 2k$